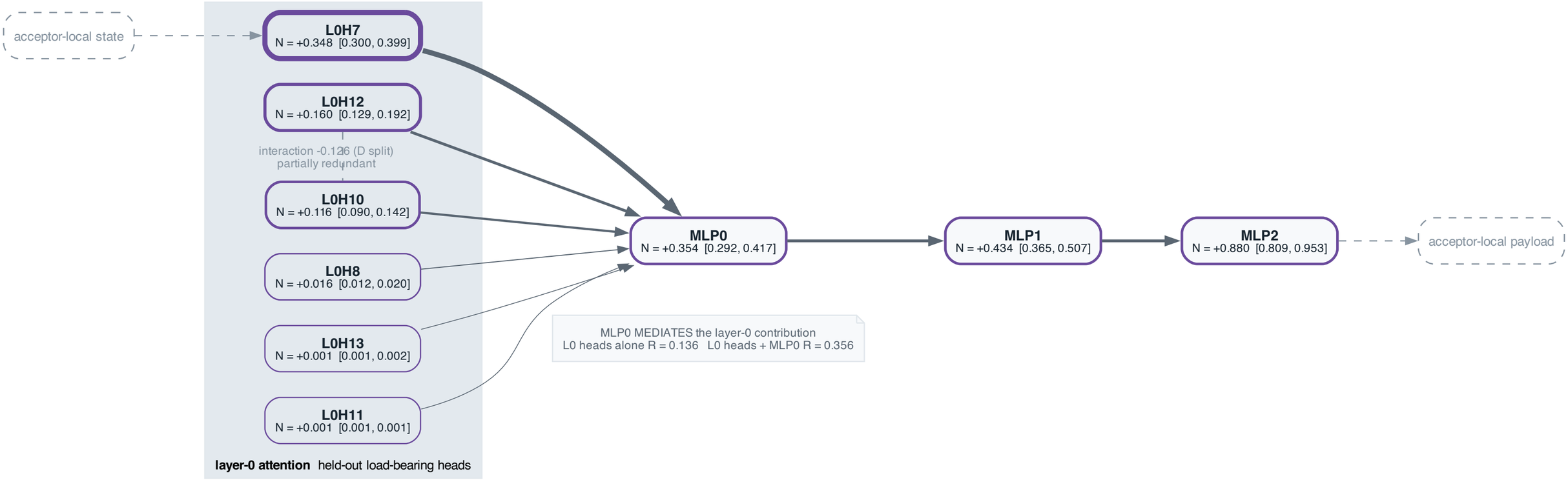

A10 / C31 payload generator, held-out (n = 220 pairs, 196 genes). N = conditional causal necessity with a gene-bootstrap 95% CI.
STRICT SERIAL chain: each MLP is necessary on held-out, none is sufficient alone (MLP0 only 0.015, MLP1 only 0.010, MLP0+MLP1 0.005).
The N values are CONDITIONAL effects. They do not sum and are not additive fractions.
Generator ceiling 0.282. Value path only: the address is held clean, so this graph shows how the payload is BUILT, not how it is addressed or delivered.
Dashed grey boxes are scope context, not measured nodes.